Price Dynamics for Credence Products: Penetration and Skimming Strategies

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Abstract

This paper investigates a dynamic model of price signaling for a credence product of unknown social performance. A new entrant uses prices to reveal social responsibility and uphold reputability throughout a verification phase. In equilibrium, the signaling strategy involves penetration or skimming pricing depending on the competitive pressure faced by the entrant. Confronted with competition from conventional incumbents, the socially responsible entrant repeatedly charges low prices to penetrate the market. By contrast, in an untapped market, the socially responsible entrant repeatedly charges high prices to skim the cream off the top of the demand. In both cases, costly signaling is consistent with Veblen's law that conspicuous waste is an effective signal of reputability.

Keywords: Credence product, Penetration pricing, Signaling, Skimming pricing, Social responsibility.

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1 Introduction

As Baron (2011) points out, some firms voluntarily enrich their products or services with credence attributes to differentiate them from their conventional counterparts. These credence attributes commit the firm to fulfilling pro-social objectives, such as the sustainable management of natural resources and environmental protection (organically produced food, low-emissions electricity), support for fair trade relationships or attention to animal welfare (free-range poultry). Credence attributes are so valuable to consumers that they are willing to pay a price premium for them. However, a credence attribute is not easily observable to consumers. To make purchase decisions, consumers have to rely on information provided by experts, mostly through certification. Certification mitigates the adverse selection problem[1] raised by credence attributes, as long as it is trustworthy and free of conflicts of interest. Nevertheless, certification is itself a credence service. If consumers are not sure that certification conveys the truth, they are still not able to evaluate whether information disclosed by the certifying body is accurate or not after purchase of the credence product. Even repeat purchases and use of the product provide little information on the environmental harm or the labor exploitation resulting from the firm's operating practices.

The growing practices of greenwashing (Delmas and Burbano, 2011, Lyon and Maxwell, 2011) or false advertising (Rhodes and Wilson, 2018) provide evidence that certification of environmental responsibility is not a panacea

[1] Adverse selection is the fraudulent behavior characterized by people taking advantage of their private information about some hidden attributes. Adverse selection was identified by Akerlof (1970) in the markets for used cars with hidden flaws ("lemons").

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for adverse selection. For instance, a number of European car makers recently cheated consumers in commercials promoting "Clean Diesel" vehicles as environmentally friendly (Bachmann et al., 2017).[2] Besides fraudulent labelling of environmental attributes, the credibility of certification is a challenge for other dimensions of social responsibility. Greenpeace and the Environmental Investigation Agency have produced evidence that the Forest Stewardship Council certification was granted to logging companies operating with little regard for sustainability or even legality.[3] The same issue arose with Fair Trade certification[4] or the Marine Stewardship Council.[5] Incidentally, in all these examples, learning about social responsibility built on investigations conducted by non-governmental organizations.

Besides certification, there are manifold means of communicating information about social performance to consumers, such as the release of assessment reports, investigation findings, quality reviews, comparative advertising, rating by agencies or test results. All these mechanisms require a significant

[2] On 18 September 2015, the United States Environmental Protection Agency accused German automaker Volkswagen Group to breach the Clean Air Act. This charge was based on a study on nitrogen oxide (NOx) emissions discrepancies between European and US models of vehicles commissioned in 2014 by the International Council on Clean Transportation. This independent nonprofit organization found that Volkswagen had equipped diesel cars with software designed to cheat on emissions tests. The fraud allowed the vehicles' NOx output to meet US standards during regulatory testing but emit up to 40 times more NOx in real-world driving conditions. Thereafter, independent tests carried out by the German car club ADAC, the world's second-biggest motoring organization, proved that, under normal driving conditions, diesel vehicles including the Volvo S60, Renault's Espace Energy and the Jeep Renegade, exceeded legal European emission limits for NOx by more than 10 times. In January 2017, Volkswagen pleaded guilty to criminal charges.
https://www.theguardian.com/business/ng-interactive/2015/sep/23/volkswagen-emissions-scandal-explained-diesel-cars
https://www.independent.co.uk/news/business/news/volkswagen-emissions-scandal-more-carmakers-implicated-tests-reveal-pollution-levels-popular-diesels-a6674386.html.

[3] https://e360.yale.edu/features/greenwashed-timber-how-sustainable-forest-certification-has-failed.

[4] Weitzman, Hal (9 September 2006). "The bitter cost of Fair Trade coffee": https://www.ft.com/content/d191adbc-3f4d-11db-a37c-0000779e2340

[5] Jacquet et al. (2010) raises concern about potential conflicts of interest involved in the certification of sustainable fisheries.

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time frame to extract private information from the firm and make it publicly available. Inevitably, after the introduction of a product or service claimed to be socially responsible, there is a *verification phase*, the end of which comes when the initially private information becomes public, and the duration of which depends on the operating speed and efficiency of the means deployed to verify and spread information. Before private information becomes public, price signaling is arguably the fastest way of credibly revealing information.

Although uninformed buyers cannot learn the social performance of a product through consumption, they can infer it from observing the firm's decisions in the short run, such as pricing (provided that the firm has some market power). Usually, consumers make repeat purchases before complete information is available. Throughout the verification phase, the firm may perfectly signal social responsibility to consumers every time it posts a price. Then, the credibility required for the price signal, which is inherently part of a separating Bayesian equilibrium, guarantees the credibility of certification momentarily until the next price signal. Regarding credence products, the credibility of price signaling is a dynamic issue that builds on the repetition of price signals. Even though one-shot signaling reveals the truth about the firm's social performance at the moment, consumers' belief is challenged next time by the threat of misleading pricing which persists throughout the verification phase.

The purpose of this paper is to examine the credibility of price signaling in the market for credence products. For this, we investigate a dynamic model of price signaling whose structure and equilibrium concept borrow from the works of Bagwell and Riordan (1986), Milgrom and Roberts (1986), Vincent (1998) and, more recently, Kaya (2009). In the present model, an entrant introduces a new product with a "socially responsible" label that indicates a quality beyond the conventional standard (with no social concern). Initially, social performance is the entrant's private information. Hence, consumers are not able to assess by themselves the true social responsibility or to judge

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whether information provided by the label is accurate or not. Making public the firm's private information through investigation and tests requires a time lag. This verification phase involves a number of repeat purchases. Every time consumers purchase the product, they try to infer from the price whether the product is responsible or conventional. The entrant may reveal information through separating prices in any period of the verification phase.

One-shot separation can be achieved in a given period before full information is publicly available provided that, first, the responsible entrant finds it advantageous to be perceived as responsible rather than conventional, and second, the conventional entrant fails to masquerade as the responsible one. Even though one-shot signaling reveals full information about social responsibility, it does not establish the firm's reputability once and for all. The responsible entrant must still continue to convince consumers that she is indeed responsible because the conventional entrant's motive for cheating consumers remains unchanged until full information is publicly available. Defending reputability is an unremitting challenge for the responsible entrant during the verification phase.

We examine two marketing strategies of penetration pricing and skimming pricing. We show that both of them support separation along the equilibrium path of the dynamic signaling game under two scenarios that differ according to the competitive pressure faced by the entrant: the scenario of the occupied market where incumbents are competing to offer the conventional product and the scenario of the untapped market for the credence product.

Penetration pricing is a pricing strategy where the responsible product is introduced and remains throughout the verification phase at a low price, which subsequently rises to the higher price of complete information. As a result, penetration pricing is an efficient means of signaling social responsibility when the entrant is facing competition from conventional incumbents. This is consistent with the lesson from the marketing literature that penetration pricing arises mostly in highly competitive markets (Spann et al.,

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2015). The least costly signaling price at which penetration pricing achieves separation is the market price for the conventional product. We also characterize the minimum value for the discount factor above which separation is an equilibrium. Hence, sales prospect are critical to signaling the entrant's social responsibility when the market is already occupied by conventional incumbents. Moreover, a higher level of social responsibility or more severe competition in the occupied market facilitate separation.

Skimming pricing strategy involves charging a high introduction price which remains unchanged during the verification phase and declines to the complete information level. It turns out that skimming pricing is an efficient means of signaling social responsibility in an untapped market. The reason is that the conventional entrant finds it more costly to drive consumers away with high prices than does the responsible entrant. This temporary potlatch is necessary to demonstrate social responsibility and defend the entrant's reputability. Again, this is consistent with the marketing principle that firms with significant market power or strong brand reputation are more likely to follow a skimming strategy (Spann et al., 2015).

Both pricing strategies entail losses of profits for the responsible entrant during the verification phase. Compared to the complete information benchmark, the introduction price is distorted downward with penetration pricing and upward with skimming pricing. To deter the conventional entrant from mimicking, the responsible entrant must distort price repeatedly before full information is publicly available, thereby ostentatiously sacrificing profit. Upon observing this, consumers get convinced that the entrant is socially responsible and their belief remains unshakable until full information is publicly available. This is another instance of Veblen's law that conspicuous waste is an effective signal of reputability.

The finding that skimming pricing achieves separation in an untapped market extends to credence products the rule of high and declining prices established by Bagwell and Riordan (1991) for experience products. In con-

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trast, it may seem surprising that low prices prove to be a credible signal of social responsibility when the market is already occupied. However, in a market where competition between conventional incumbents pushes down prices, profit margins are low, which provides a conventional entrant with a strong incentive to trick consumers into buying at high prices. Therefore, low prices are more reliable than high prices to prevent mimicry and send a credible signal of social responsibility. As a result, low prices achieve separation by attracting consumers away from the established competitors. Moreover, low prices maintain separation throughout the whole verification phase, at the price of conspicuous losses. Thus, a low and rising price path identifies a responsible product when the market is occupied by conventional firms.

The rest of this paper is organized as follows. Section 2 discusses a review of the related literature. Sections 3-4 present the one-shot model under complete information. Section 5 investigates the existence of separating equilibria in the dynamic game with repeated price signaling. In Section 6, we fully characterize the equilibrium outcome that involves penetration pricing in the scenario of the occupied market. In Section 7, we show that skimming pricing achieves separation in the scenario of the untapped market. Our concluding remarks appear in Section 8.

2 Related literature

One-shot signaling game has proven to be well suited to studying the pricing of experience products.[6] The huge literature on the subject emphasizes separating equilibrium rather than pooling equilibrium. One reason may be that given in Mahenc (2017): separating prices demonstrate that certification is trustworthy, whereas pooling prices invalidate the credibility of certification. Another reason is that firms sometimes prefer to communicate private infor-

[6] Nelson (1970) coined the term "experience" for attributes that are not determined prior to purchase but can be revealed through consumption.

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mation about product quality indirectly through price signaling rather than directly through voluntary disclosure (Daughety and Reinganum, 2008, in a monopoly setting; Janssen and Roy, 2015, in a duopoly setting). Finally, a third reason comes from Cho and Kreps (1987). They provide persuasive arguments to consider that the most reasonable outcome of a game with one signaling period is separating rather than pooling: the "intuitive criterion" eliminates all but the most efficient separating equilibrium outcome, known as the "Riley" outcome.

Darby and Karni (1973) initiate the idea that credence attributes may not be evaluated even after purchase and consumption. Accordingly, we consider here that the time lag in obtaining complete information on social responsibility, called the verification phase, is spread over a number of trading periods that can be finite or infinite. The repetition of the trading relationship during the verification phase allows consumers to observe more than one period's price and use the information conveyed by past prices to make inferences and form beliefs. The firm's reputability builds on the history of earlier price signaling and the responsible firm maintains her reputability with the help of current price signaling.

It is instructive to relate this dynamic analysis to theoretical literature on the price dynamics for experience goods. In a pioneering article on firms' provision of quality in a market for an experience good, Wolinsky (1983) demonstrates the existence of a separating price equilibrium, in which each price signal exceeds the marginal cost of producing the quality it signals. Thereafter, a number of theoretical articles have examined the dynamic path of price signaling for experience goods (Bagwell, 1990; Bagwell and Riordan, 1991; Judd and Riordan, 1994). A result common to most of these papers is that a high introductory price signals high quality successfully from the beginning of the trading relationship. Skimming pricing for high quality is the efficient means of revealing information because an upward distortion in price is needed to deter opportunistic firms from mimicking price at the out-

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set. In the present model, the upward distortion arises for the same reason when the market is untapped, and must be repeated in every period of the verification phase to uphold reputability. In Bagwell and Riordan (1991), the signaling distortion depends on the ratio of informed to uninformed consumers, which increases over time. As time passes, prices signaling high quality decline because the signaling distortions diminish as information diffuses. Judd and Riordan (1987) agree that prices should decline as consumers gain more product-specific information through experience.

Bagwell (1990) argues for penetration rather than skimming pricing to signal high quality in the markets for experience products. He investigates a two-period signaling model in which an entrant introduces an experience product into a market where an incumbent firm is selling a low-quality product. This matches the scenario of the occupied market in the present paper. Bagwell shows that separation cannot occur in the first period. There only exists pooling equilibria in which entrants with different qualities charge the same price, so that consumers cannot initially make inferences as to the entrant's true quality. The introductory price for high quality is set below the production cost to attract consumers. This is fairly similar to the equilibrium outcome of our dynamic game in the scenario of the occupied market: the penetration pricing strategy entails losses of profits throughout the verification phase. The main difference with the pooling price outcome in Bagwell (1990) is that signaling distorts downward the price of the responsible product both to reveal that the entrant's quality is high and to uphold her reputability thereafter.

In markets for environmentally friendly goods, Mahenc (2007 and 2008) and Sengupta (2012) find that a monopolist may need to distort price upward in order to signal high environmental quality unless it is cheaper to produce than low environmental quality (in which case the price distortion is downward).

Despite the great variety of credence goods and services (Balafoutas and

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Kerschbamer, 2020), there is little empirical work on the price dynamics or on the signaling role of prices in credence goods markets. One exception is the market for pharmaceutical drugs where the use of promotional prices strongly resembles the penetration strategy: prices are kept low early in the life cycle of a branded drug before rising after its patent expires (Frank and Salkever, 1997; Bhattacharya and Vogt, 2003). Bhattacharya and Vogt (2003) argue that such a price dynamics serves to build public knowledge about the drug.

A stylized fact of the economic literature is that the move toward more responsible products and services should command a price premium. However, the studies that examine the relationship between responsible production or certification and price in the markets for ecological products draw mixed conclusions. Some organic products receive price premiums above their conventional counterparts[7], while some other sell at lower prices relative to comparable conventional products.[8]

In practice, penetration pricing strategies can be observed within the organic and natural foods sector: Costco and Kroger, two major grocery store chains in the US, use market penetration pricing for the organic foods they sell. Another example of penetration pricing is carbon pricing. Boyce (2018) finds that most actually existing carbon prices fall significantly short of the levels recommended by climate policy analysts. The implementation of a climate policy via a tax rate indexed to a fixed emission reduction target can be seen as a credence service. If the policymaker's environmental friendliness is private information, citizens are uncertain about the effectiveness of the climate policy; for instance, citizens do not observe how much the policymaker's preferences diverge from the Paris Agreement's objective of holding the rise

[7] Casadesus-Masanell et al. (2009) for organic cotton shirts compared to regular cotton; Kiesel et al.(2005) for organic milk and milk that is free of genetically modified recombinant Bovine Growth Hormone.

[8] Kiesel et al. (2005) for unlabeled milk that is free of genetically modified recombinant Bovine Growth Hormone; Delmas and Grant (2014) for California organic wines advertising their certification on their labels; Abraben et al. (2017) for organic Tuscan red wine with high quality ratings.

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in global mean temperature to 1.5-2 °C above pre-industrial levels. The carbon tax is the price paid by citizens for the policy implementation. The reason why existing carbon prices are so low may be that pro-environmental policymakers use them to get the environmental policy early accepted and adopted by citizen with the hope of keeping them once prices will rise to higher Paris-consistent levels. The temporary failure to comply with the target standards is an effective signal of environmental reputability.

3 The model

Consider the market for a conventional product or service that generates a negative externality affecting the environment or the society. As it causes discomfort to consumers, this externality undermines the utility of the product. For instance, consumers suffer from health problems related to polluting emissions or they feel guilty for contributing to the damage caused by the adverse effects of production. The total number of consumers is normalized to unity. Consumers derive the same gross surplus of value v for the conventional product. Each consumer purchases at most one unit of the product that generates the amount $1-e$ of harmful by-product, where $e \in [0,1]$ is an indicator of social performance for the product. Social performance includes environmental protection as well as ethical, political or religious concerns. For the conventional product, the social performance is $e = 0$.

Consumers feel different distastes x for the negative externality: a consumer with distaste x bears a utility loss $(1-e)\,x$. Measured in monetary terms, this loss may represent spending on health care or adaptation costs to deal with a polluted environment. The parameter x is uniformly distributed along a segment of unit length. Hence, consumers distinct from 0 partly internalize the negative externality. Another interpretation is that distaste heterogeneity reflects the degree of social conscience of consumers. If, for instance, the good is fossil energy, consumers may differ in their dislike of

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the negative impact on global warming, and if it is nuclear energy, they may differ in their dislike of the potential risks imposed on future generations by nuclear repositories.

A consumer with distaste x who purchases one unit of the product with social performance e at price p has the following indirect utility

$$u = v - (1 - e)\, x - p \tag{1}$$

An entrant designated by the pronoun "she", intends to introduce a new variant of the product with credence attributes that cannot be observed for some time, possibly an infinite length of time. These attributes represent social benefits, whose level is measured by e. For example, if the credence attributes enhance environmental quality, e is the pollution abatement. In the remainder of the paper, the firm whose product complies with the standard e is referred to as being socially responsible. According to (1), consumers are ready to pay more for a responsible product than for its conventional counterpart.

Compared to conventional production, the commitment to social responsibility raises the overall cost of production. There is a fixed setup cost $F > 0$: in addition to the fee paid to have the product certified, F includes unsalvageable expenditures in developing the social attribute and specific entrepreneurial skills, sunk investments, for instance in specialized machinery for green production and fuel-saving equipment, or long-term rental contracts that cannot be resold. Moreover, compliance with the standard e entails a constant unit cost $c(e)$, with $c'(e) > 0$. Let $\varphi = c(e) - c(0)$ represent the difference in costs of using responsible versus conventional methods.

The decision to enter the market is made more or less difficult depending on whether the market is untapped or occupied. Therefore, our analysis deals with two scenarios: in the first scenario, the entrant breaks into an occupied market where perfectly competitive incumbents sell the conventional product at the market price $p_c = c(0)$; in the second scenario, the entrant enters an

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untapped market. In the occupied market, the entrant differentiates her product from the conventional one, while she is faced with no competition in the untapped market.

Let $\widetilde{x}$ denote the critical value of the taste distribution that determines market demands.

In the occupied market, the distaste $\widetilde{x}$ of the consumer indifferent between purchasing the conventional product and the responsible product sold at price p is given by

$$v - \widetilde{x} - p_c = v - (1 - e)\,\widetilde{x} - p \tag{2}$$

All consumers with tastes x that satisfy $x \geq \frac{p-p_c}{e}$ purchase the responsible product and the remaining consumers purchase the conventional product. This implies the following demand function for the responsible product

$$D(p, e) = 1 - \widetilde{x} = \begin{cases} 1 \text{ if } p \leq p_c \\ 1 - \frac{p-p_c}{e} \text{ if } p_c < p \leq e + p_c \\ 0 \text{ if } p \geq e + p_c \end{cases} \tag{3}$$

Note that the price-elasticity of demand is

$$\eta\,(p) = -\frac{\partial D\,(p, e)}{\partial p}\frac{p}{D\,(p, e)} = \frac{p}{e + p - p_c} \tag{4}$$

The more responsible the product or the more intense competition between conventional incumbents, the less demand for the responsible product is sensitive to price.

In the untapped market, consumer $\widetilde{x}$ is indifferent between purchasing and not purchasing the responsible product

$$0 = v - (1 - e)\,\widetilde{x} - p \tag{5}$$

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Then the demand function is

$$D(p,e)=\widetilde{x}=\begin{cases} 1 \text{ if } p\leq v+e-1 \\ \frac{v-p}{1-e} \text{ if } v+e-1<p\leq v \\ 0 \text{ if } p\geq v \end{cases} \tag{6}$$

The demand for the responsible product when it is sold at marginal cost is

$$D(c\,(e)\,,e)=\begin{cases} 1-\frac{c(e)-p_c}{e} \text{ in the occupied market} \\ \frac{v-c(e)}{1-e} \text{ in the untapped market} \end{cases} \tag{7}$$

To ensure that the responsible product is socially desirable in both scenarios, we assume

$$c\,(e)<\min\{e+p_c,v\} \tag{8}$$

4 Complete information

When information is complete, the firm who achieves social performance at a level e earns the operational profit $\pi_r\,(p)=(p-c(e))D(p,e)$. The expression of the profit-maximizing price $\widehat{p}_r$ depends on whether the market is occupied or untapped

$$\widehat{p}_r=\begin{cases} \frac{e+p_c+c(e)}{2} \text{ in the occupied market} \\ \frac{v+c(e)}{2} \text{ in the untapped market} \end{cases} \tag{9}$$

At that price, demand is positive when the product of social performance e is socially desirable. Indeed, when (8) holds,

$$D(\widehat{p}_r,e)=\begin{cases} \frac{e+p_c-c(e)}{2e} \text{ in the occupied market} \\ \frac{v-c(e)}{2(1-e)} \text{ in the untapped market} \end{cases}>0 \tag{10}$$

Under complete information, the maximized profit $\widehat{\pi}_r$ is given by

$$\widehat{\pi}_r=\pi_r\,(\widehat{p}_r)-F=\begin{cases} \frac{(e-\varphi)^2}{4e}-F \text{ in the occupied market} \\ \frac{(v-c(e))^2}{4(1-e)}-F \text{ in the untapped market} \end{cases} \tag{11}$$

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In the scenario of the occupied market, the entrant whose social performance is $e = 0$ is faced with competitive incumbents selling the conventional product at the market price $p_c = c(0)$. At such a low price, the conventional entrant cannot afford the certification fee and thus stays out. We will use the convention that the price for the conventional product under complete information is

$$\widehat{p}_c = \begin{cases} p_c \text{ in the occupied market} \\ \frac{v+c(0)}{2} \text{ in the untapped market} \end{cases} \tag{12}$$

Proposition 1: *Suppose that* (8) *holds, i. e. the responsible product is socially desirable. Under complete information, an entrant finds it profitable to introduce a responsible product, provided that*

- $F \leq \frac{(e-\varphi)^2}{4e}$ *when there is competition from conventional incumbents;*
- $F \leq \frac{(v-c(e))^2}{4(1-e)}$ *when there is no competition.*

5 The dynamic signaling game

Our purpose is to explore the dynamics of price signaling in a market for credence products whose attributes remain unobservable to consumers for some time but then become public knowledge. During the lag time before social performance is observed, a conventional firm may cheat on consumers by mimicking the price of the responsible product. Thus, the responsible entrant must defend her reputability every period before full information is publicly available.

Initially, consumers cannot ascertain whether the new product is conventional (indexed by c) or responsible (indexed by r). Let $\Theta = \{c, r\}$ be the set of possible types for the entrant and her product. It is commonly

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known that the new product is responsible with probability μ_0 and conventional with probability $1-\mu_0$. It takes T periods of observation (verification and communication) before consumers learn the social performance of the new product, where $T \leq +\infty$. Social responsibility is the entrant's private information during consumer verification phase.

The entrant and consumers play an infinitely repeated game in which $\delta \in (0,1)$ is the discount parameter per period for each player. Time is discrete and dates are denoted by $t = 0, 1, 2, ...$ We call dynamic game the repeated game obtained by repeating T times the one-shot signaling game. We assume that the price $p_{\theta t}$ charged by the entrant of type θ in period t is perfectly observed by all consumers.

A sequence of prices charged by the responsible entrant up to period $t \leq T$ defines a history in period t which will be denoted by $h^t = \{p_{r1}, p_{r2}, ..., p_{rt}\}$. If consumers have concluded period $t-1$ with the probability μ_{t-1} that the new product is responsible, the entrant enters period t with a reputation μ_{t-1}. We use the Markov refinement that μ_{t-1} is a sufficient statistic for h^{t-1} and contains all the relevant information needed by the entrant to make optimal decisions. Hence, the price chosen by the entrant in period t contingent on the history h^{t-1} will depend only on μ_{t-1}. Let $\mu_t(.) : \mathbb{R}_+ \times [0,1] \rightarrow [0,1]$ be consumers' posterior belief which updates prior belief μ_{t-1} with the observation of p_{rt}.

If $\mu = \mu(p)$ is the probability assigned to the true type being responsible after observing the price p, we define $D(p,\mu)$ to be the demand for the new product under incomplete information

$$D(p,\mu) = \begin{cases} 1 - \frac{p-p_c}{\mu e} \text{ in the occupied market} \\ \frac{v-p}{1-\mu e} \text{ in the untapped market} \end{cases} \tag{13}$$

The profit made by the entrant of type θ will be $\pi_\theta(p,\mu)$. Let $\widehat{p}_\theta(\mu)$ denote the optimal price that maximizes $\pi_\theta(p,\mu)$, yielding $\widehat{\pi}_\theta(\mu) = \pi_\theta(\widehat{p}_\theta(\mu), \mu)$.

We restrict attention to pure strategies. A pure strategy for the entrant

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of type θ is a sequence of prices and beliefs $\left(\{\sigma_{\theta t}\}_{t=1}^{T}, \theta \in \Theta, \{\mu_t\}_{t=1}^{T}\right)$, where the function $\sigma_{\theta t} : [0,1] \times \Theta \to \mathbb{R}_+$ assigns the price $p_{\theta t}$ chosen by the type θ in period t given the prior belief μ_{t-1} (summarizing h^{t-1}) .

We look for pure strategy perfect Bayesian equilibria (PBE). The solution concept of PBE requires that the sequence of signaling strategies and the consumers' system of beliefs satisfy both sequential rationality and consistency. Sequential rationality imposes that, after every history up to a given period, the responsible entrant chooses price so as to maximize her discounted sum of profits given the consumers' system of beliefs. Consistency requires that consumers use Bayes' rule to update their beliefs whenever possible. We find the solution recursively, starting with period T, the last period before full information is publicly available. The entrant enters period $t \leq T$ with a reputation μ_{t-1} and chooses her best price, given consumers' purchase decisions and the impact of current behavior on the next period's reputation. The probability μ_{t-1} is then revised in light of the price observed on the equilibrium path according to Bayes' rule. Off the equilibrium path, Bayes' rule does not apply since information sets are reached with probability zero. As usual, this freedom in specifying off-equilibrium path beliefs leads to a multiplicity of equilibria.

We begin by characterizing the set of separating equilibria. In the dynamic signaling game, "separating" means that the equilibrium is separating in every period of the verification phase. For a sequence of price signals $\{p_{\theta t}\}_{t=1}^{T}$ to be on the path of a separating equilibrium, it must satisfy individual rationality (IR) constraints and incentive compatibility (IC) constraints.

The IR constraints ensure that the responsible entrant is willing to continue revealing its true social performance at every moment of the verification phase

$$\forall t \leq T : \sum\nolimits_{k=t}^{T} \delta^{k-t} \pi_r(p_{rk}, 1) + \frac{\delta^{T+1-t}}{1-\delta} \widehat{\pi}_r \geq \frac{1}{1-\delta} \widehat{\pi}_r(0) \tag{14}$$

The left-hand side of (14) is the expected value of the stream of future

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profits made by the responsible entrant along the equilibrium path from period t onwards, discounted to period t. The right-hand side is the best continuation profit following a deviation that the responsible entrant can guarantee herself.

The IC constraints guarantee that, for each point in time, the conventional entrant does not mimic the responsible entrant from this point until full information is publicly available

$$\forall t \leq T : \sum\nolimits_{k=t}^{T} \delta^{k-t} \pi_c(p_{rk}, 1) + \frac{\delta^{T-t+1}}{1-\delta} \widehat{\pi}_c \leq \frac{1}{1-\delta} \widehat{\pi}_c \tag{15}$$

The left-hand side is the expected value of the stream of future profits made by the conventional entrant when she masquerades as the responsible entrant from period t until full information is publicly available. The right-hand side is the expected value of the stream of profits obtained by the conventional entrant along the equilibrium path.

Consider the case $T = 1$. The new product is an experience product since consumers need only one consumption period to learn the true social performance. Any deviation in period 1 can affect only current profits since the entrant has fixed strategies in all future periods and future consumers perfectly know her type. The game reduces to the one-shot signaling game with initial belief μ_0. Obviously, the entrant charges the complete information price $\widehat{p}_c$ to signal the conventional type, because it is the best price regardless of consumer beliefs.

Suppose that the entrant charges p_1 to signal the responsible type. From (15), the conventional type should not be tempted to mimic p_1. For $T = 1$, this constraint boils down to

$$\pi_c(p_1, 1) \leq \widehat{\pi}_c \tag{16}$$

We can expect a whole continuum of separating equilibria to exist, provided that the equality version of (16) admits at least one solution, which

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supports the Riley outcome. Let $p_{\mathcal{R}}$ denote this price. In the Riley outcome, the conventional entrant is indifferent between revealing its actual type and charging $p_{\mathcal{R}}$ to mimic the responsible type; that is,

$$\pi_c\left(p_{\mathcal{R}}, 1\right) = \widehat{\pi}_c \tag{17}$$

Hence, the price $p_{\mathcal{R}}$ signals social responsibility in the least costly manner. Formally, for any p_1 such that $\pi_c\left(p_1, 1\right) < \widehat{\pi}_c$, we have

$$\pi_r\left(p_1, 1\right) + \frac{\delta}{1-\delta}\widehat{\pi}_r < \pi_r\left(p_{\mathcal{R}}, 1\right) + \frac{\delta}{1-\delta}\widehat{\pi}_r \tag{18}$$

The intuitive criterion (Cho and Kreps, 1987) places additional restrictions on out-of-equilibrium beliefs, which singles out $p_{\mathcal{R}}$ as the most reasonable equilibrium outcome among all the separating equilibria. According to this refinement, if consumers observe $p_{\mathcal{R}}$ off the equilibrium path instead of an equilibrium price p_1 such that $\pi_c\left(p_1, 1\right) < \widehat{\pi}_c$, they should be convinced that the entrant is responsible with a probability of 1 at $p_{\mathcal{R}}$. This is because the conventional entrant has no incentive to undertake such a deviation regardless of consumer beliefs, whereas the responsible entrant could increase her profits if consumers were putting probability 1 to the responsible type upon seeing $p_{\mathcal{R}}$. This implicit message makes the deviation worthwhile for the responsible entrant and the equilibrium price p_1 is discarded by the intuitive criterion. As a result, the equilibrium price $p_{\mathcal{R}}$ is the only one to be robust to the intuitive criterion.

If the responsible entrant chooses to deviate from $p_{\mathcal{R}}$, one step is missing in the time needed to learn about social performance. Then, the lowest continuation profit the entrant can guarantee herself is $\widehat{\pi}_r\left(0\right)$ from period 1 onwards. The entrant is willing to signal social responsibility via $p_{\mathcal{R}}$ if this price satisfies

$$\pi_r\left(p_{\mathcal{R}}, 1\right) + \frac{\delta}{1-\delta}\widehat{\pi}_r \geq \frac{1}{1-\delta}\widehat{\pi}_r\left(0\right) \tag{19}$$

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This condition is necessary for the existence of a separating equilibrium in the one-shot signaling game. Suppose for a contradiction that there exists a separating equilibrium supported by p_1 different from $p_{\mathcal{R}}$, while (19) does not hold. Then, inequality (18) holds, which implies that

$$\pi_r(p_1, 1) + \frac{\delta}{1-\delta}\widehat{\pi}_r < \frac{1}{1-\delta}\widehat{\pi}_r(0) \tag{20}$$

Thus, the responsible entrant would prefer to deviate towards $\widehat{p}_r(0)$ rather than signaling her true type even if $\widehat{p}_r(0)$ will forever convince consumers that she is conventional. This contradicts the proposition that p_1 supports an equilibrium.

Lemma 1: *Consider the case $T = 1$ of an experience product and suppose that* (19) *holds. There exists a continuum of separating equilibria among which that supported by $p_{\mathcal{R}}$ is the least costly way of signaling social responsibility successfully.*

The existence of the Riley outcome as an equilibrium ensures that separation is possible in the one-shot signaling game. In the dynamic signaling game, we look for solutions recursively, starting with period T. At the beginning of period T, consumers believe that the product is responsible with probability μ_{T-1}. Taking this probability to be a sufficient statistic for the history h^{T-1}, we can think of period T as a one-shot signaling game with initial belief μ_{T-1}. Again, the Riley outcome deserves special attention since separation cannot be possible on the equilibrium path of the dynamic game if it fails at the end of the verification phase.

The problem of multiplicity of equilibria encountered in the one-shot signaling game is exacerbated in the dynamic signaling game. To deal with this issue, we require that separating price signaling be efficient in continuation equilibrium and so we focus attention on the price signal $p_{\mathcal{R}}$. Henceforth, our goal is to show that the T-time repetition of the Riley outcome is a separating PBE of the dynamic game.

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Suppose that the sequence of prices $\{p_{rt}\}_{t=1}^{T}$ signals the responsible type on the path of a separating equilibrium. It is useful to consider a dynamic process that describes how strategies and beliefs are computed at every node in the game tree. For some period $t \leq T$, we define strategies as follows

$$\sigma_c^1(\mu_0) = \widehat{p}_c \text{ and } \sigma_r^1(\mu_0) = p_{\mathcal{R}} \tag{21}$$

$$\begin{aligned} \sigma_c^t(\mu_{t-1}) &= \widehat{p}_c \text{ for all } \mu^{t-1} \\ \sigma_r^t(\mu_{t-1}) &= \begin{cases} p_{\mathcal{R}} \text{ if } \mu_{t-1} = 1 \\ \text{and } \widehat{p}_r(0) \text{ otherwise} \end{cases} \end{aligned} \tag{22}$$

and beliefs as follows

$$\mu_{t-1} = \begin{cases} 1 \text{ if } p_{rt-1} = p_{\mathcal{R}} \\ \text{and } 0 \text{ otherwise} \end{cases} \tag{23}$$

The dynamic process given by (22) implies that the responsible entrant plays the following trigger strategy: "reveal your actual type in the least costly manner if all previous signals have been fully informative, otherwise switch forever to the best outcome you can guarantee yourself". The entrant enters period t with reputation μ_{t-1}. The system of beliefs (23) satisfies consistency: Bayes' rule applies when the responsible entrant charges the equilibrium separating price $p_{rt-1} = p_{\mathcal{R}}$, therefore updated belief is $\mu_{t-1} = \mathbf{1}$. Moreover, any deviation from the equilibrium path ($p_{rt-1} \neq p_{\mathcal{R}}$) ruins the entrant's reputation irreversibly. The deviation is viewed by consumers as "proof" that the entrant is conventional and, from (22), consumers are unshakable in this conviction once it is formed. This is a common restriction imposed on equilibria in multiperiod signaling games. Osborne and Rubinstein (1990) refer to this restriction as "never dissuaded once convinced": once consumers are convinced at some point in the game that the entrant is

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of the conventional type with a probability of 1, subsequent actions will not alter such a strong belief.

Lemma 1 shows that strategies (22) and beliefs (23) achieve a PBE for $T = 1$ provided that (19) holds. The same argument also applies for $t = T$ and thus the least costly way of signaling social responsibility is to charge $p_{rT} = p_{\mathcal{R}}$. Furthermore, the intuitive criterion selects $p_{rT} = p_{\mathcal{R}}$ as the unique equilibrium price path beginning at T. Moving back another period, strategies (22) and beliefs (23) achieve a PBE for $T - 1$, which is the the unique equilibrium price path robust to the intuitive criterion beginning at $T - 1$. Proposition 2 below follows by a recursive argument.

Suppose that strategies (22) and beliefs (23) can be supported as PBE following some period $t + 1 < T$. In period t, the entrant charges the full information price $\widehat{p}_c$ to signal the conventional type, regardless of the reputation. If μ_{t-1}=1, the entrant enjoys the best possible reputation when she enters period t. Charging $p_{\mathcal{R}}$ to masquerade as the responsible type before full information is publicly available, the conventional entrant makes the following profit value discounted to period t:

$$\sum\nolimits_{k=t}^{T} \delta^{k-t} \pi_c(p_{\mathcal{R}}, 1) + \frac{\delta^{T-t+1}}{1-\delta} \widehat{\pi}_c = \frac{1}{1-\delta} \widehat{\pi}_c \tag{24}$$

Thus, the conventional entrant is not tempted to mimic $p_{\mathcal{R}}$, and hence condition (15) is met.

If the responsible entrant has ruined her reputability by deviating from $p_{\mathcal{R}}$ in period $t-1$, she enters period t with the worst possible reputation μ_{t-1}=0. Consumers' conviction that the entrant is conventional cannot later on be updated to assign positive probability to the responsible type. As $T-t$ steps are missing in the time needed to learn about social responsibility, consumers will always mistake the responsible type for the conventional one. Therefore, the lowest continuation profit the responsible entrant can guarantee herself is $\widehat{\pi}_r(0)$ from period t onwards. From (14), the responsible entrant is willing

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to signal her true social performance in period t if the price $p_{\mathcal{R}}$ satisfies

$$\pi_r\left(p_{\mathcal{R}},1\right)\left(1+\delta+...+\delta^{T-t}\right)+\frac{\delta^{T-t+1}}{1-\delta}\widehat{\pi}_r\geq\frac{1}{1-\delta}\widehat{\pi}_r\left(0\right) \tag{25}$$

Since the IR condition (25) must hold for every period of the verification phase, the most demanding condition occurs when $t=1$; that is,

$$\pi_r\left(p_{\mathcal{R}},1\right)\left(1+\delta+...+\delta^{T-1}\right)+\frac{\delta^{T}}{1-\delta}\widehat{\pi}_r\geq\frac{1}{1-\delta}\widehat{\pi}_r\left(0\right) \tag{26}$$

Proposition 2: *There exists a sequence of price signals* $\{p_t\}_{t=1}^{T}$ *such that* $p_t=p_{\mathcal{R}}$ *signals social responsibility successfully along the equilibrium path if and only if* (26) *holds.*

Condition (26) can be rewritten as follows

$$\delta^T\left(\widehat{\pi}_r-\pi_r\left(p_{\mathcal{R}},1\right)\right)\geq\widehat{\pi}_r\left(0\right)-\pi_r\left(p_{\mathcal{R}},1\right) \tag{27}$$

The solution in T for the equality version of (27) is the maximum time lag in making public the information about social responsibility, that is required to support separation. One can interpret δ^T as the speed of making public the firm's private information: when T rises, information communication takes a longer time and δ^T falls. For separation to occur, it must be that the entrant is sufficiently patient or the information communication is fast enough.

Before full information is publicly available, the entrant must squander enough money to convince consumers that she is responsible. The left-hand side of (27) represents the cost of preventing cheating, that is, the spending of resources needed to build up and protect reputability during the verification phase. This conspicuous spending must be greater than the foregone profit resulting from consumer misperception given in the right-hand side of (27).

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Costly signaling follows Veblen's law that conspicuous waste is an effective signal of reputability.

6 Penetration pricing

In this section, we focus on the scenario in which the entrant is faced with competitive incumbents selling the conventional product at the market price p_c. So would do the entrant were she perceived as being herself conventional. In that case, the entrant could not afford if only the certification fee, so the entrant stays out regardless of her actual type. We follow the convention that the price $\widehat{p}_c = p_c$ signals the conventional entrant in every period along the separating equilibrium path and results in the equilibrium profit $\widehat{\pi}_c$=0. Similarly, $\widehat{\pi}_r(0)$=0 is the best profit the responsible entrant can get when consumers are certain that she is conventional.

Suppose that p_r credibly signals the responsible type in equilibrium of the one-shot signaling game. Then, the IC constraint (16) requires

$$(p_r - c(0))\left(1 - \frac{p_r - p_c}{e}\right) \leq 0 \tag{28}$$

To reveal the truth, the responsible entrant must distort price either downward ($p_r \leq c(0)$) or upward ($p_r \geq p_c + e$) relative to the full information price $\widehat{p}_r = \frac{e+p_c+c(e)}{2}$. Were distortion upward, prices would be so high that no consumer would purchase the responsible product. Obviously, neither learning nor reputation building will take place if there is no consumption. Thus, the putative separating prices p_r are downward distorted and bounded above by $p_{\mathcal{R}} = c(0)$, that is, the price supporting the Riley outcome in the present scenario: the entrant sets this price at the same level as p_c, the market price for the conventional product.

In the scenario of the occupied market, the IR condition (26) can be

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rewritten as

$$\frac{\delta^T}{1-\delta}\frac{(e-\varphi)^2}{4e} \geq \varphi\frac{1-\delta^T}{1-\delta} \tag{29}$$

To signal truthfully social responsibility, the entrant uses a penetration pricing strategy: she enters and charges the low price p_c in every period $t \leq T$; afterwards, full information is publicly available and the price rises up to the complete information level $\widehat{p}_r$. Clearly, this strategy requires the responsible entrant to sacrifice profits during the verification phase. The low price is necessary to make consumers aware of the new product, attract them away from the conventional competitors and build market share for the responsible product. The claim that the new product is responsible enhances consumers' willingness-to-pay. Together with this claim, the price p_c switches all consumers to the responsible product and capture them during the whole verification phase. What makes both the claim and the price credible is the conspicuous loss of profits incurred by the entrant, which is represented by the right-hand side of (29). Penetration pricing can achieve separation if and only if condition (29) is met. This condition says that the flow of profits accruing from the sales of the responsible product after the verification phase more than offsets the loss of profits during the verification phase. As consumer switching is not worth as much to the conventional entrant, she is never tempted to mimic the penetration price, and so she is staying out.

Proposition 3: *Suppose that the market is occupied,* $c(e) < e + p_c$*, i. e. the responsible product is socially desirable, and* (29) *holds. Then, the entrant can credibly signal social responsibility with the penetration strategy to charge* $p_{\mathcal{R}} = p_c$ *repeatedly throughout the verification phase and subsequently raise the price toward* $\widehat{p}_r$*.*

The signal repeatedly sent by penetration pricing during the verification phase helps the entrant build up and uphold the reputability of social responsibility. The message to consumers is : "There is nothing to lose to purchase the new product since it is as cheap as the conventional product. Try it out

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and you will be unshakably convinced that it is responsible. To prove this, I am switching all consumers away from the conventional competitors, which makes me lose money throughout the verification phase. Everyone knows that I would not make such a gift were my product conventional."

Condition (29) is equivalent to

$$\delta \geq \widetilde{\delta}^{\frac{1}{T}}, \tag{30}$$

where $\widetilde{\delta} = \frac{4e\varphi}{(e+\varphi)^2}$. Hence, $\widetilde{\delta}^{\frac{1}{T}}$ is the lowest discount factor for which separation occurs. The incentive to deter cheating comes through sales prospect.

Notice that $\widetilde{\delta}^{\frac{1}{T}}$ is increasing in T and approaches 1 as T goes to infinity. That is, separation is more difficult to sustain, requiring a more patient entrant (or less frequently changed prices) when the verification phase lasts longer. Moreover, $\widetilde{\delta}$ is decreasing in both the degree of differentiation e and the cost advantage of the conventional method φ, thereby increasing in p_c: more differentiation through social performance or stiffer competition in the occupied market facilitates separation. The intuition is that the demand curve for the responsible product is less elastic when there is more differentiation between the responsible and the conventional products or when competition is more intense between conventional incumbents. The less elastic demand allows the responsible entrant to extract more consumer surplus in the long run when private information becomes public. Higher discounted future returns provide the responsible entrant with a stronger incentive to signal her type.

7 Skimming pricing

In this section, we turn to the scenario in which the market is untapped so that the entrant enjoys a monopoly position. In every period along the separating equilibrium path, the conventional entrant signals her true social

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performance with the complete information price $\widehat{p}_c = \frac{v+c(0)}{2}$, thereby making the profit $\widehat{\pi}_c = \frac{(v-c(0))^2}{4}$. Moreover, the responsible entrant signals her type truthfully throughout the verification phase by charging the price $p_{\mathcal{R}}$ repeatedly from period 1 till period T. Since consumers are convinced that the entrant is conventional after observing a deviation from $p_{\mathcal{R}}$, the best profit the responsible entrant can make in that case is $\widehat{\pi}_r(0) = \frac{(v-c(e))^2}{4}$ by charging $\widehat{p}_r(0)$. To guarantee that the price signal $p_{\mathcal{R}}$ achieves separation from the very beginning of the verification phase, the IR condition (26) requires

$$(p_{\mathcal{R}} - c(e)) \left(\frac{v - p_{\mathcal{R}}}{1-e}\right) \frac{1-\delta^T}{1-\delta} + \frac{\delta^T}{1-\delta} \frac{(v-c(e))^2}{4(1-e)} \geq \frac{1}{1-\delta} \frac{(v-c(e))^2}{4} \quad (31)$$

The set of prices for which (31) holds is the interval $P_r = \left[\underline{p}_r, \overline{p}_r\right]$ where

$$\underline{p}_r = \widehat{p}_r - \frac{(v-c(e))\sqrt{e\left(1-\delta^T\right)}}{2\left(1-\delta^T\right)} \quad (32)$$

and

$$\overline{p}_r = \widehat{p}_r + \frac{(v-c(e))\sqrt{e\left(1-\delta^T\right)}}{2\left(1-\delta^T\right)} \quad (33)$$

The responsible entrant would prefer to deviate to $\widehat{p}_r(0)$ rather than revealing her type with a price outside P_r. In other words, the price $p_{\mathcal{R}}$ must belong to P_r to uphold the reputability of social responsibility throughout the whole verification phase.

We have previously seen that $p_{\mathcal{R}}$ solves equation (17). In the present scenario, substituting p for $p_{\mathcal{R}}$ into (17) yields

$$(p - c(0)) \left(\frac{v-p}{1-e}\right) = \frac{(v-c(0))^2}{4} \quad (34)$$

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This equation has an upper and lower root

$$\overline{p}_c = \widehat{p}_c + \frac{(v - c(0))\sqrt{e}}{2} \tag{35}$$

and

$$\underline{p}_c = \widehat{p}_c - \frac{(v - c(0))\sqrt{e}}{2} \tag{36}$$

The price $p_{\mathcal{R}}$ cannot be inside $\left(\underline{p}_c, \overline{p}_c\right)$, otherwise the conventional entrant would be tempted to mimic $p_{\mathcal{R}}$. Calculations show that $\underline{p}_c \notin P_r$ while $\overline{p}_c \in P_r$. Indeed, we have

$$\underline{p}_c - \underline{p}_r = \frac{1}{2}\left(\varphi - (v - c(e))\sqrt{e}\left(\frac{\sqrt{(1-\delta^T)}}{(1-\delta^T)} + 1\right)\right) < 0 \tag{37}$$

and

$$\overline{p}_r - \overline{p}_c = \frac{1}{2}\left(\varphi + (v - c(e))\sqrt{e}\left(\frac{\sqrt{(1-\delta^T)}}{(1-\delta^T)} - 1\right)\right) > 0 \tag{38}$$

It is straightforward that the expression in (37) is negative. The expression in (38) is positive because it is an increasing function of δ^T, which is positive when $\delta = 0$. Finally, the only possible signals the responsible entrant can send to reveal her type are prices above $\overline{p}_c$. This is because when the entrant raises price above $\widehat{p}_r$, the cost of driving consumers away is higher for the conventional type than it is for the responsible type. As a result, $p_{\mathcal{R}} = \max\{\widehat{p}_r, \overline{p}_c\}$ is the least costly way of signaling social responsibility. It turns out that $p_{\mathcal{R}} = \overline{p}_c > \widehat{p}_r$ when $\varphi < (v - c(0))\sqrt{e}$: in that case, the entrant must distort price upward relative to the complete information price.

Proposition 4: *Suppose that the market is untapped and $c(e) < v$, i. e. the responsible product is socially desirable. Then, for all δ, i.e. no matter how patient the entrant, a separating equilibrium exists where the repetition of the price $p_{\mathcal{R}} = \max\{\widehat{p}_r, \overline{p}_c\}$ signals social responsibility.*

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Furthermore, if $c(0) + \frac{\varphi}{\sqrt{e}} < v$*, then the entrant can credibly signal social responsibility with the skimming strategy to charge* $p_{\mathcal{R}} = \overline{p}_c$ *repeatedly throughout the verification phase and subsequently bring the price down to* $\widehat{p}_r$*.*

Unlike the scenario of the occupied market, both types of entrant have sales prospect under complete information. Therefore, the possibility of signaling social responsibility does not depend on sales prospect. The assumption that the responsible product is socially desirable is sufficient for the existence of a separating equilibrium.

If the difference in costs of using responsible versus conventional methods is small enough, then the entrant charges the high price $\overline{p}_c$ in every period of the verification phase to signal social responsibility. As soon as full information is publicly available, the price drops down to the complete information level $\widehat{p}_r$. The initial high price reduces demand compared to the complete information regime, thereby skimming the cream of the market that is the most willing to pay for social responsibility. This conspicuous price distortion signals that the entrant can afford to squander a specific sales volume. Actually, this is the minimum quantity of sales that the conventional entrant cannot forgo. Therefore, consumers correctly infer from observing $\overline{p}_c$ that the entrant is responsible.

Price skimming enables the responsible entrant to establish her reputability from the first period of the verification phase. Although information is revealed at the outset, the responsible entrant continues to distort the price upward along the equilibrium path until full information is publicly available. To uphold reputability throughout the verification phase, the responsible entrant needs to deter mimicry in every period. Consumers are paying too high a price for an unremitting demonstration of social responsibility. This is another instance where Veblen's law of conspicuous waste for the sake of reputability applies.

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8 Conclusion

This paper has examined the dynamic path of price signaling for a product with a new attribute of social responsibility. As social responsibility is a credence attribute, it can hardly be observed by consumers through experience or use during a verification phase that includes a number of repeat purchases. Before full information becomes publicly available, price signaling is the only source of information for consumers. We have tackled this issue by means of a dynamic signaling game.

We show that the dynamic signaling game possesses a separating equilibrium in which the least costly signaling strategy in the one-shot signaling game reveals information in every period of the verification phase. The competitive pressure from conventional incumbents crucially affects the pricing pattern along the equilibrium path. In the presence of conventional incumbents, separation requires a penetration strategy; that is, a downward distorted price is needed to signal the responsible product throughout the verification phase. By contrast, separation requires a skimming strategy in an untapped market; that is, an upward distorted price is needed to signal the responsible product throughout the verification phase.

Both strategies of penetration and skimming pricing require the responsible firm to squander money in every period of the verification phase. The resulting loss of profits deters opportunistic mimicry, thereby proving to consumers that the new product is responsible. Indeed, faced with competitive incumbents, the conventional entrant has no incentive to charge the low price of the penetration strategy. Similarly, the conventional entrant could not afford the loss in sales volume entailed by the high price of the skimming strategy. In both scenarios, wasteful signaling follows the same logic as Veblen's law that conspicuous waste is an effective signal of reputability.

Although the price distortion reveals information from the beginning, the new entrant must continue to convince consumers that she is indeed socially responsible by distorting the price over and over throughout the verification

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phase. Once built, reputability must be upheld. It is not possible to shift all the signaling cost to earlier periods of the verification phase because the conventional type of entrant would choose to imitate the responsible type if this one were charging the complete information price in one period or another of the verification phase.

Our results suggest that price signaling is likely to be easier and more effective for a socially responsible firm in an untapped market than in a market occupied by conventional competitors. In the former scenario, the skimming strategy achieves separation no matter how fast full information about social responsibility is made publicly available. In the latter scenario, the penetration strategy reveals information in a credible manner provided that turning private information into public information does not take too long. Moreover, the price signaling strategy is easier to implement when the demand curve for the responsible product is less elastic. There may be two reasons for this: social performance differentiates more the product from its conventional counterpart or there is stiffer competition between conventional incumbents.

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